

HCP Profiling App • Territory Reconfiguration Web Portal

Executive Implementation Blueprint: Dedicated Web App Reconfiguration Engine, Hideable Sidebar & Live ERPNext Tree Sync

Platform: Dedicated Web App (Desktop / Laptop Browser) **Access Point:** HCP Mobile App Menu Link (Role-Gated: Admin & SFE Only)

Navigation: Hideable Left Sidebar • Clean Pure-Text Controls • Dropdown Cycle Vault **Backend Core:** ERPNext v15 REST API (Nested Set Traversal & Atomic Primary Key Cascade)

1. Clean Architecture: Pure Text UI, Hideable Sidebar & Tree-Gated Security

Pure-Text Typography & Hideable Sidebar

CLEAN UX

- Zero Unwanted Images/Emojis:** Buttons and navigation use clean, high-clarity text labels without distracting icons.
- Collapsible Left Sidebar:** The 5 core reconfiguration tabs are housed in a hideable left navigation bar, maximizing screen space for wide multi-column tables.
- Repositioned Actions:** "Sample CSV Template" and "Refresh Live Sync" moved out of the center banner into dedicated header and sidebar action docks.

Live Tree Gating & Role Isolation

ADMIN & SFE ONLY

- Tree-Verified Programs Only:** Selection is restricted strictly to programs active in both Territory and Sales Person trees (Abbott DC, Bayer, RiteMed, Vivaro).
- Zero Dummy Accounts:** Only real doctor accounts from ERPNext `HCP Account` and real rep nodes from `Sales Person` are rendered.
- Blank Historical Cycle Policy:** Previous cycle columns (Cycle 1–3) remain blank (–) until actual cycle Excel files are staged.

2. Interactive UI Layout: Modern Hideable Sidebar Workspace

HCP RECONFIG PORTAL • Program: `ABBOTT DIABETES CARE` • Cycle: `CYCLE 4 (PROPOSED)` Actions: `SAMPLE CSV` `REFRESH SYNC` • User: `admin@pims.com` [SFE Lead]

NAVIGATION [HIDE --]

Pre-Import CSV Preview

Program Code Renaming

Multi-Cycle Historical Matcher

Realignment & Archive Canvas

Pre-Flight Simulation & Cutover

STATUS: `LIVE MASTER CONNECTED`

ACTIVE COMMERCIAL BASELINE (ABBOTT DC TERRITORIES) **23 REAL TERRITORIES LINKED**

Hist Code	Proposed Code	District Group	Assigned Rep	Docs	Status
AD0101	AD0101	AD1 - GMA/NORTH LUZON	GRAZIEL RIVO	0 Docs	<code>UNCHANGED</code>
AD0102	AD0102-NEW	AD1 - GMA/NORTH LUZON	LIZA ROXAS	3 Docs	<code>RENAMED</code>
AD0104	AD0104	AD2 - GMA/SOUTH LUZON	VINA TAMAYO	2 Docs	<code>UNCHANGED</code>
AD0106	AD0106	AD0105 (COOR)	JONEL APSAY	2 Docs	<code>UNCHANGED</code>

3. Six-Step End-to-End Operational Workflow for SFE Leads

STEP 1

LAUNCH

Desktop Browser Access via Mobile Launcher Screen

ROLE PROTECTED

- Admin/SFE taps "Territory Reconfiguration" in mobile app drawer; app starts local HTTP server on port 8765.
- Launcher screen displays instant copyable short URLs and fast-scanning QR code for desktop browsers on local Wi-Fi.

STEP 2

PREVIEW

File Ingestion & Live Master Verification Grid

FILE INTENT CHECK

- Portal automatically hydrates live active territories and doctor counts directly from ERPNext masterlist.
- Optional CSV upload allows staging proposal files; file intent confirmation verifies syntax and encoding before processing.

STEP 3

RENAME

Program Code Renaming Studio & Cascade Matrix

PRIMARY KEY SAFE

- SFE selects territory code and assigns new restructured code; system stages atomic primary key rename.
- Live cascade matrix immediately displays affected doctor accounts and downstream document links.

STEP 4

MATCH

Clean Dropdown Cycle Vault & Multi-Cycle Matcher

BLANK CYCLES

- Dropdown selector replaces cluttered buttons; displays real doctors from ERPNext `HCP Account`.
- Historical cycle columns are blank (–) pending Excel upload, ensuring zero fabricated historical claims.

STEP 5

CANVAS

Consolidated District Canvas & Dual-Tree Alignment

MANPOWER REALLOC

- Consolidates territories into downsized district clusters; decommissioned codes safely move to Archive node.
- Dual-Tree visualizer verifies exact 1:1 synchronization between Territory Tree and Sales Person Tree without orphan nodes.

STEP 6

CUTOVER

Pre-Flight Simulation & Live Database Cutover

LIVE COMMIT

- Simulates foreign key integrity, verifies 0 circular loops, and recalculates nested set boundaries.
- Commits changes live to ERPNext; mobile app sync reflects updated roster on next refresh without reinstall.

Accurate Tree Synchronization Architecture: Territory & Sales Person Trees

Technical Engine Mechanics: How ERPNext Synchronizes Primary Keys, Nested Sets & Manpower Absorptions Live

DocTypes Modified: Territory • Sales Person • HCP Account • HCP Profile Submission

Algorithm: Modified Preorder Tree Traversal (Nested Set 'lft'/'rgt' Re-indexing)

Integrity Method: 'frappe.rename_doc' Atomic Cascading Foreign Key Transactions

Data Loss: Absolute Zero • Decommissioned Nodes Archived In-Place

4. Detailed Synchronization Mechanics: What Happens When Reconfiguration Executes?

When the SFE Lead clicks **"Confirm & Execute Live Reconfiguration Now"**, the Web App initiates a deterministic, transactional 6-phase synchronization protocol with ERPNext:

Phase 1: Primary Key Renaming ('frappe.rename_doc')

PRIMARY KEY CASCADE

- Atomic Database Cascade:** When a territory code is renamed (e.g. `RM101` → `ADC-T01`), Frappe executes an engine-level rename across all relational tables.
- Zero Orphan References:** Every linked record in `HCP Account.territory`, `HCP Profile Submission.territory`, `Customer`, and `Sales Invoice` is updated in the same SQL transaction.
- Audit Logging:** A permanent rename log is recorded in ERPNext DocType `Route History` and version audit tables.

Phase 2: Nested Set Recalculation ('rebuild_tree')

LFT / RGT INTEGRITY

- Modified Preorder Tree Traversal:** ERPNext represents `Territory` as a Nested Set with integer boundary coordinates (`lft` , `rgt`).
- Atomic Tree Re-indexing:** When territories are re-parented into consolidated district clusters (e.g. `AD1 - GMA/NORTH LUZON`), Frappe invokes `NestedSet.rebuild_tree("Territory", "parent_territory")`.
- Mathematical Loop Prevention:** Guarantees 0 circular references, maintains leaf-to-root validation, and ensures tree traversal queries execute in sub-millisecond time.

Phase 3: Sales Person Hierarchy Absorption

MANPOWER SYNC

- Surviving Representative Allocation:** When headcount is reduced, consolidated child territories are reassigned to the active representative in `Sales Person` tree.
- Parent Hierarchy Sync:** Re-parents field rep nodes under surviving District Sales Managers (e.g., Raymond Viray for BA1, Regalado Banasihan for AD1, Miriam Quinan for RiteMed).
- Zero Unassigned Territory:** Ensures every reconfigured territory node has an active manager and user ID link, preventing orphaned accounts.

Phase 4: In-Place Doctor Account Re-Linking

DOCTOR ROSTER

- In-Place Updates:** The web app executes `frappe.db.set_value` on affected `HCP Account` records, updating `territory` and `sales_person`.
- Zero Duplicate Doctors:** Doctor profiles in masterlist `HCP` are completely untouched; only the program-specific affiliation in `HCP Account` reflects the new territory code.
- Continuous Compliance:** Specialization, workplace affiliations, and contact records remain 100% intact.

Phase 5: Zero-Deletion Archival Policy

ARCHIVE NODE

- Non-Destructive Relocation:** Decommissioned territories are re-parented under `Archived Territories` (`is_group: 0`) instead of being deleted.
- Historical Audit Continuity:** Preserves past interaction logs, call metrics, and historical profiling submissions for audit defense.
- Instant Headcount Restoration:** If manpower expands in future quarters, archived territories can be restored to active clusters with a single click.

Phase 6: Live Cutover & Mobile Synchronization

MOBILE BROADCAST

- Cache Invalidation:** Embedded `TerritoryWebServer` broadcasts sync invalidation signals to active mobile app sessions.
- Real-Time Field Reflection:** On next app refresh or login, mobile MedReps immediately see updated territory assignments without reinstalling the app.
- Offline Safety:** The mobile app's local SQLite database seamlessly merges updated territory keys, preventing data collisions.

5. Exact ERPNext REST APIs Powering the Tree Synchronization Engine

POST `/api/method/frappe.client.rename_doc`

PRIMARY KEY CASCADE

- Payload:** `{"doctype": "Territory", "old_name": "AD0102", "new_name": "AD0102-NEW", "merge": 0}`
- Cascades primary key rename across all linked HCP Accounts, submissions, and sales records atomically.

PUT `/api/resource/HCP Account/{id}`

IN-PLACE DOCTOR SYNC

- Payload:** `{"territory": "AD0102-NEW", "sales_person": "LIZA ROXAS (AD0102)"}`
- Updates doctor territory and rep link in-place with zero duplicate records created in masterlist.

PUT `/api/resource/Territory/{id}`

RE-PARENT & CLUSTER

- Payload:** `{"parent_territory": "AD1 - GMA/NORTH LUZON", "territory_manager": "LIZA ROXAS"}`
- Re-parents child node into consolidated district and triggers automated nested set boundary recalculation.

GET `/api/live-data?program={name}`

LOCAL WEBSERVER

- Served by embedded `TerritoryWebServer` on port 8765; provides instant real-time synchronization with zero ERPN Desk dependency.

6. Implementation & Operational Rollout Governance

- Phase 1 (Portal UI & Sidebar Overhaul):** Deploy clean pure-text portal with hideable left navigation sidebar and dropdown cycle vault.
- Phase 2 (Live Tree Integration):** Enforce strict tree-gated programs (Abbott DC, Bayer, RiteMed, Vivaro) with zero dummy accounts.
- Phase 3 (Dry-Run Simulation):** Execute pre-flight verification checks to confirm zero circular loops and 100% referential integrity.
- Phase 4 (Live National Cutover):** Broadcast reconfigured territory hierarchy live to mobile field representatives across the country.